

Functions, Derivatives & Partial Derivatives:

1. Function:

a machine : input x in $\rightarrow$ output y out



Simple Example:

$$\begin{aligned} f(x) &= x^2 \\ \Rightarrow f(2) &= 4; \\ f(3) &= 9; \\ f(-1) &= 1; \end{aligned}$$

DL Example:

$$\begin{aligned} f(x) &= W \cdot x + b \\ &\text{(func. of a single neuron)} \end{aligned}$$

2. Derivatives:

$$x + 0.0001$$

$f(x) \rightarrow$ how much change?

- measure rate of change
- nudge the input a tiny bit, how much does output move?
- that ratio is called derivative (also called slope)

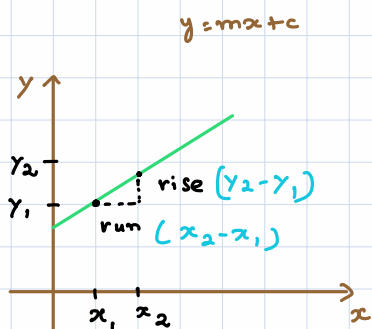
$f(x) \rightarrow$ function

$f'(x) \rightarrow$ derivative of the function

$$f'(x) = \frac{dy}{dx}$$

$dy \rightarrow$ change in y

$dx \rightarrow$ change in x



Important

Power rule:

$$f(x) = x^n$$

$$f'(x) = \frac{d}{dx} (x^n) = n \cdot x^{n-1}$$

$$\frac{d}{dx} (\text{const}) = 0$$

$$x^3 \rightarrow 3x^2$$

$$x^2 \rightarrow 2x$$

$$x \rightarrow 1$$

$$\text{const} \rightarrow 0$$

3. Partial Derivatives:

If a function has many inputs, nudge one & freeze the rest

$$f(x_1, x_2) = x_1^2 + 3x_1x_2 + x_2^2$$

$$\frac{\partial f}{\partial x_1} \quad (x_2 \rightarrow \text{const.})$$

$$\begin{aligned} x_1^2 &\rightarrow 2x_1 \\ 3x_1x_2 &\rightarrow 3x_2 \\ x_2^2 &\rightarrow 0 \end{aligned}$$

$$\frac{\partial f}{\partial x_1} = 2x_1 + 3x_2$$

$$\frac{\partial f}{\partial x_2} \quad (x_1 \rightarrow \text{const.})$$

$$\begin{aligned} x_1^2 &\rightarrow 0 \\ 3x_1x_2 &\rightarrow 3x_1 \\ x_2^2 &\rightarrow 2x_2 \end{aligned}$$

$$\frac{\partial f}{\partial x_2} = 3x_1 + 2x_2$$

Why this matters for Deep Learning?

→ understand change in Loss "L" when we change weight "w"

→ Loss "L" depends on millions of weights : $w_1, w_2 \dots w_n$

→ $\frac{\partial L}{\partial w_1}$ = how much L change when we tweak only w_1

→ Now, combine all partials together → gradient ∇L